



FIRST SEMESTER 2026-2027

Course Handout (Part II)

In addition to Part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course:

COURSE NO : **CS G623**
COURSE TITLE : **ADVANCED OPERATING SYSTEMS**
Instructor in-charge : **Asish Bera** [asish.bera@pilani.bits-pilani.ac.in]

1. COURSE OBJECTIVE

To introduce the design and implementation issues of distributed operating systems

2. PRE-REQUISITES

It is assumed that the students have done a basic course on operating systems and computer networks. Also the students should have sufficient skills in C/Java/Python programming languages.

3. SCOPE

Distributed Operating Systems work in an environment where we have independent machines (both hardware and software) connected with each other over a network. A Distributed OS makes a Distributed System a virtual uniprocessor system. The distributed OS to be studied in this course is microkernel based. It's just that the user level processes that are separated from the kernel can run on remote machines. Also, the OS has some sort of idea what machines are hooked up, and can make decisions about where to run things based on the relative speed of the machines.

4. BOOKS

Text Book

[T1] M. Singhal & N. Shivaratri, "Advanced Concepts in Operating Systems: Distributed, Database and Multiprocessor Operating Systems", Tata McGraw Hill, 2015.

Reference Books

[R1] Ajay D. Kshemkalyani, and Mukesh Singhal "Distributed Computing: Principles, Algorithms, and Systems", Cambridge University Press, 2008, (Reprint 2013).

[R2] Distributed Operating Systems – The Logical Design by A. Goscinski, AW

[R3] Distributed Systems-Concepts and Design by G. Coulouris, AW

[R4] Distributed Systems Principles and Paradigms 2/E by A. S. Tanenbaum, PHI

5. PLAN OF STUDY

Lect #	Lecture Session	Learning Outcome	REFS (Text)
1-2	Overview of Advanced Operating System, Review of concepts	Design approaches, Motivation, Review, Types of Advanced OS.	T1. Ch 1, R1. Ch 1, R2. Ch 1
3-4	Architecture and Models for Communication	Architectural Issues, Communication Networks, and Communication Primitives. Need for a model, Message Passing for IPC, Shared Variables, Relationship among models.	T1. Chap 4, R2. Ch 3



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5-7	Time in distributed systems	Limitations of DS, logical and vector clocks, physical clock synchronization, causal ordering of messages, global state, cuts of distributed computation	T1. Ch 5, R1. Ch 6, R2. 6 and 8
8-12	Distributed Mutual Exclusion	Lamport, Recart-Agrawala, and Maekawa's algorithms; Suzuki-kasami broadcast algorithm, and Raymond's tree-based algorithm	T1. Ch 6, R2. Ch 7
13-16	Global State Collection and Distributed Deadlock Detection	Resource and Communication deadlock, Strategies to handle deadlock, Deadlock Detection Algorithms	T1. Ch 7, R2. Ch 9
17-20	Agreement and Consistency Protocols	System model, Classification of agreement problems, Solutions to Byzantine agreement Problems, Dolev's algorithm	T1. Ch 8
21-25	Coordination Algorithms	Leader Election, Bully algorithm, Election in rings, Consistency, Raft, and others	R2. Ch 11
26-27	Distributed File Systems	DFS architecture, HDFS, MapReduce, CHORD	T1 Ch 9, R1.18
28-32	Distributed Scheduling	Issues in Load Distribution, Components of a load distribution algorithm, Load Distribution Algorithms, Fair-share scheduling	T1. Ch 11
33-36	Fault Tolerance and recovery	Distributed Transactions, Commit Protocols, two phase commit, Classification of failures, and others	T1. Ch 12,13, R2. Ch 12
37-40	Research direction and discussions	Familiar with Recent research paradigm and applications of distributed and related systems, etc.	Research papers

EVALUATION SCHEME

Component	Duration	Nature	Weightage	Date and Time
Mid-Semester Test	90 Min	Closed book	25 %	05/10/2026 FN2
Quiz	TBA	Closed book	7 %	TBA [Class level]
Project/Assignment	TBA	Open book	30 %	TBA
Comprehensive Exam	180 Min	Partly Open (>15%) book	38%	11/12/2026 AN

NOTICES

All notices related to the course will be posted on Nalanda.

MAKE-UP

- **For make-up in Mid-Semester Test:** Makeups, in general, are allowed only in case of medical emergency. Prior permission of the IC is mandatory.
- **For make-up in Quiz and Project Assignment components:** no makeup will be given.
- **For comprehensive examination make-up:** contact AUGSD/AGSRD.

CHAMBER CONSULTATION HOUR: Wednesday, 5-6 pm and send a prior email to the instructor.

Instructor-in-Charge, CS G623

